Christian Z. Pratt, Ph.D.

Curriculum Vitae

Email	pratt@ucsb.edu
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Publications	Google Scholar

Research positions

2026 —	Postdoctoral researcher at UC Santa Barbara with Dr. Kerem Çamsarı at the Orchestrating Physics for Unconventional Systems (OPUS) Lab in the Department of Electrical and Computer Engineering and the Institute for Energy Efficiency. Investigating the co-design of novel algorithms for machine learning and their implementations in energy-efficient probabilistic computers.
2022 - 2026	Graduate student researcher at UC Davis with Dr. James P. Crutchfield in the Complexity Sciences Center and the Department of Physics and Astronomy. Investigated the fundamental physics of computing with dynamical systems theory, stochastic thermodynamics, and superconducting circuit theory. Applications include developing energy-efficient superconducting computers.
2019 - 2021	Undergraduate researcher at UC Davis working with Dr. Michael Mulhearn in the Department of Physics and Astronomy. Re-purposed smartphone camera sensors for detecting cosmic ray muons in table top experiments.

Education

2026	Ph.D. Physics, UC Davis — Research advisor: Prof. James P. Crutchfield — Thesis title: *Computing with Dynamical Energy Landscapes*
2022	M.Sc. Physics, UC Davis
2021	B.Sc. Physics, UC Davis
2019	A.A. Mathematics, San Diego Community College District

Publications

Peer reviewed

2025	**C. Z. Pratt**, K. J. Ray, and J. P. Crutchfield. Controlled erasure as a building block for universal thermodynamically-robust superconducting computing. *Chaos: An Interdisciplinary Journal for Nonlinear Science*. 10.1063/5.0227130
2025	**C. Z. Pratt**, K. J. Ray, and J. P. Crutchfield. Extracting equations of motion from superconducting circuits. *Physical Review Research*. 10.1103/PhysRevResearch.7.013014

Under review

2026	**C. Z. Pratt**, K. J. Ray, and J. P. Crutchfield. Metastable Dynamical Computing with Energy Landscapes: A Primer. arXiv:2602.11390

In prep

C. Z. Pratt, K. J. Ray and J. P. Crutchfield. Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits.

Presentations

Invited talks

2026	*Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote seminar for the Quantum Nanostructures and Nanofabrication (QNN) Group at MIT.
2026	*Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote seminar for the Orchestrating Physics for Unconventional Systems (OPUS) Lab at UC Santa Barbara.
2026	*Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote seminar for the Alternative Computing Group at National Institute for Standards and Technology (NIST).
2026	*Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote seminar for the Türeci group at Princeton University.

2026 *Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote postdoc job talk for the NanoComputing Research Lab at Eindhoven University of Technology.

2026 *Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote postdoc job talk for the Takeuchi group at Kobe University.

2026 *Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: Remote postdoc job talk for the Yang group at National University of Singapore.

2025 *Controlled Erasure as a Building Block for Universal Thermodynamically-Robust Superconducting Computing*: In-person seminar at the Information Engines at the Frontiers of Nanoscale Thermodynamics workshop in Telluride, Colorado.

2025 *Controlled Erasure as a Building Block for Universal Thermodynamically-Robust Superconducting Computing*: In-person seminar at the Molecular Foundry at LBNL.

2025 *Computing with Dynamical Energy Landscapes*: Poster presentation at the Interdisciplinary Graduate Research Exhibition at UC Davis.

2023 *Universal Dynamical Computing on the Nanoscale*: In-person seminar at the Information Engines at the Frontiers of Nanoscale Thermodynamics workshop in Telluride, Colorado.

Contributed talks

2026 *Comparing Langevin and SPICE Simulations of Dynamical Landscape Computing with Superconducting Circuits*: In-person guest seminar for the Redwood Center at UC Berkeley.

2025 *Controlled Erasure as a Building Block for Universal Thermodynamically-Robust Superconducting Computing*: 15 minute talk at Society for Industrial and Applied Mathematics (SIAM) Conference on Applications of Dynamical Systems (DS25) in Denver, Colorado.

2025 *Controlled Erasure as a Building Block for Universal Thermodynamically-Robust Superconducting Computing*: 15 minute talk at APS March Meeting 2025 in Anaheim, California.

2024 *Controlled Erasure as a Building Block for Universal Thermodynamically-Robust Superconducting Computing*: Recorded technical seminar at the Complexity Sciences Center (CSC) at UC Davis.

2024 *Universal Dynamical Computing on the Nanoscale*: 15 minute talk at Dynamic Days 2024 in Davis, California.

2023	*Universal Dynamical Computing on the Nanoscale*: 15 minute talk for a Army Research Office on-site visit at the Complexity Sciences Center (CSC) at UC Davis.

Teaching

2026	9A: Classical mechanics
2025	256A: Physics of Information 256B: Physics of Computation
2021 – 2022	7B: Fluid mechanics, electrical circuits, Newtonian mechanics 7C: Modern physics, waves, optics 9A: Classical mechanics

Awards & Honors

2025	Ryan Couch Memorial Travel Award Department of Physics and Astronomy University of California, Davis
Fall 2025, Winter 2025, Spring 2024, Spring 2023	Graduate Student Researcher Fellowship Department of Physics and Astronomy University of California, Davis
2018	NASA Community College Aerospace Scholar
2016 – 2019	Dean's Honors List San Diego Community College District

Service and Outreach

2025	Workshop Volunteer Organizer. *Physics of Agency*. Beyond Institute for Theoretical Science (BITS). Pioneer, California.
2025	Session Chair for CS28: Electronic, Optical, and Condensed Matter Systems. *Society for Industrial and Applied Mathematics (SIAM) Conference on Applications of Dynamical Systems (DS25)*. Denver, Colorado.

2020 – 2021	Mentor, Support Encourage and Develop For Children (SendForC)
2018	Volunteer, New Story Charity
2017 – 2018	Volunteer, San Diego Rescue Mission
2016 – 2018	Volunteer, San Diego Air and Space Museum

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